

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

The de Broglie wavelength (λ) of an electron can be calculated using the formula : $\lambda = h / \sqrt{2 m * K}$), where h is Plan ck's constant , m is the mass of the electron , and K is the kinetic energy .

The given kinetic energy is $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. Rearrange the formula to solve for λ : $\lambda = h / \sqrt{2 m * K}$

Substitute values , we get $\lambda = h / \sqrt{2 \times 9.11 \times 10^{-31} * 1.6 \times 10^{-19}}$

Simplify and convert to unit cm , then the de Broglie wavelength $\lambda \approx 1.23 \times 10^{-7} \text{ cm}$.

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